Assignment_Day_2

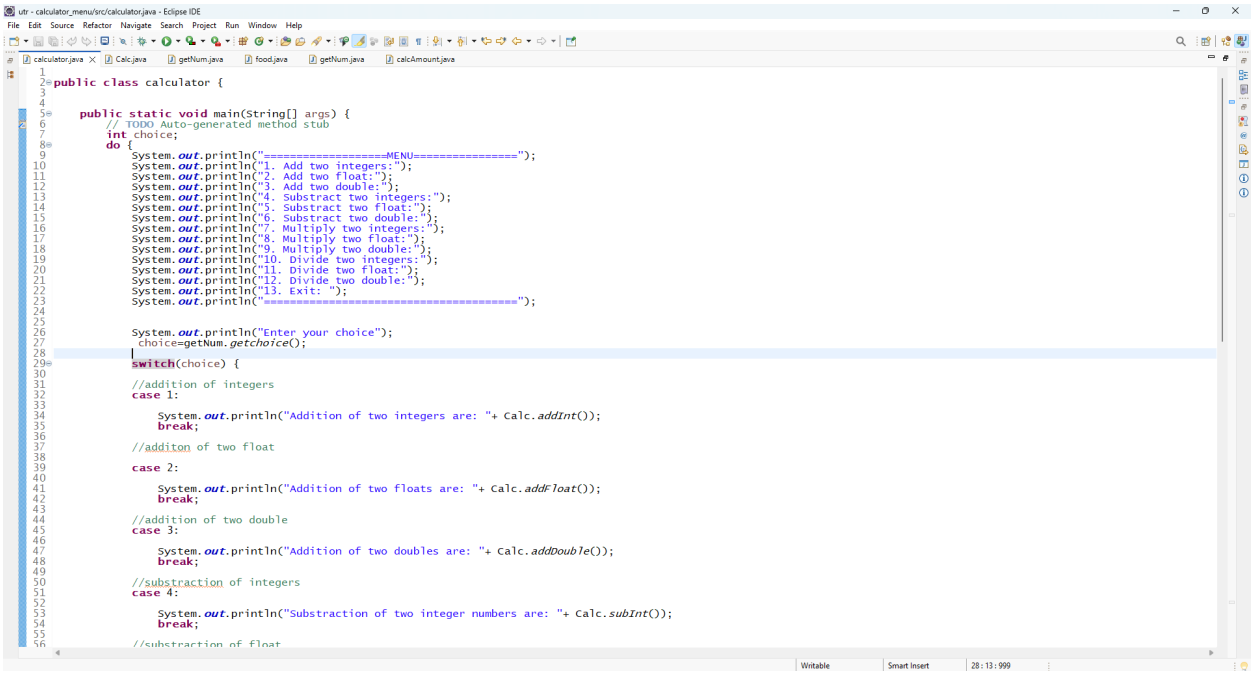
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Que 1: Create a menu driven program for calculator to add , subtract, multiply and divide two integers, two floats, two doubles.

The application will continuously display the menu till the exit choice is not selected.

File:- calculator.java



```
public class calculator {


    public static void main(String[] args) {
        // TODO Auto-generated method stub
        int choice;
        do {
            System.out.println("==================MENU===============");
            System.out.println("1. Add two integers:");
            System.out.println("2. Add two float:");
            System.out.println("3. Add two double:");
            System.out.println("4. Substract two integers:");
            System.out.println("5. Substract two float:");
            System.out.println("6. Substract two double:");
            System.out.println("7. Multiply two integers:");
            System.out.println("8. Multiply two float:");
            System.out.println("9. Multiply two double:");
            System.out.println("10. Divide two integers:");
            System.out.println("11. Divide two float:");
            System.out.println("12. Divide two double:");
            System.out.println("13. Exit: ");
            System.out.println("=====================================");


            System.out.println("Enter your choice");
             choice=getNum.getchoice();

            switch(choice) {

            //addition of integers
            case 1:

                System.out.println("Addition of two integers are: "+ Calc.addInt());
                break;

            //additon of two float

            case 2:

                System.out.println("Addition of two floats are: "+ Calc.addFloat());
                break;

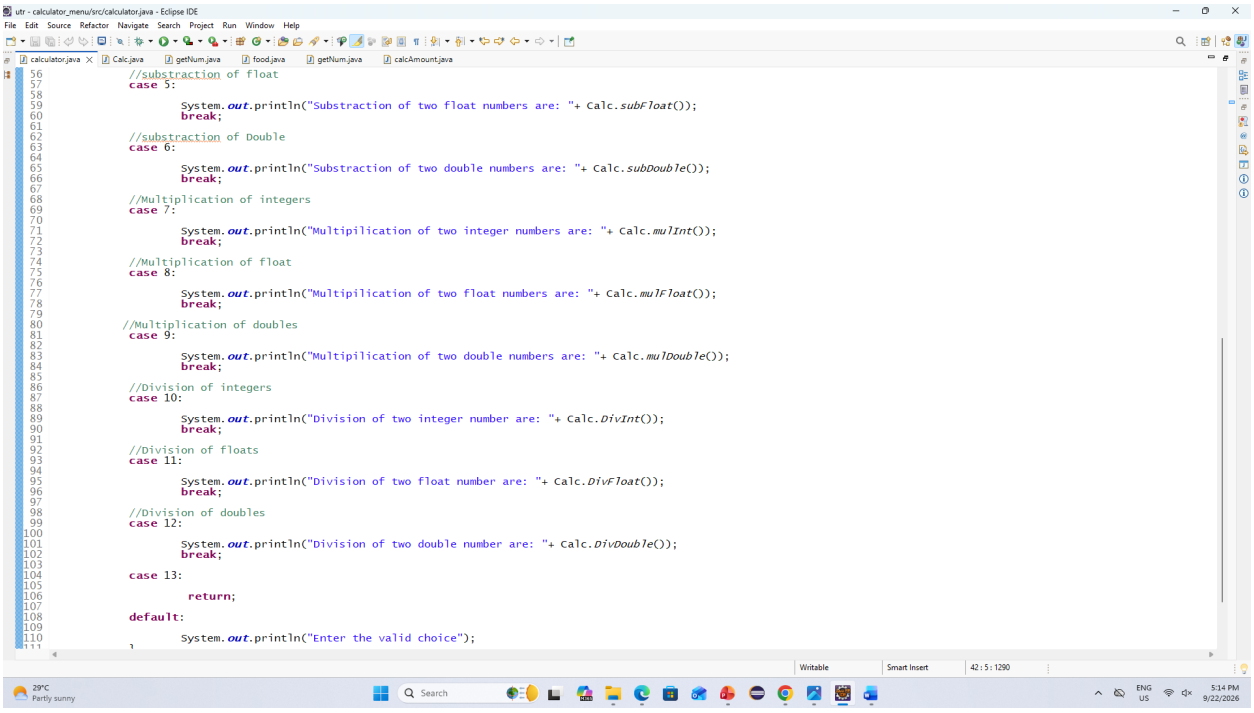
            //addition of two double
            case 3:

                System.out.println("Addition of two doubles are: "+ Calc.addDouble());
                break;

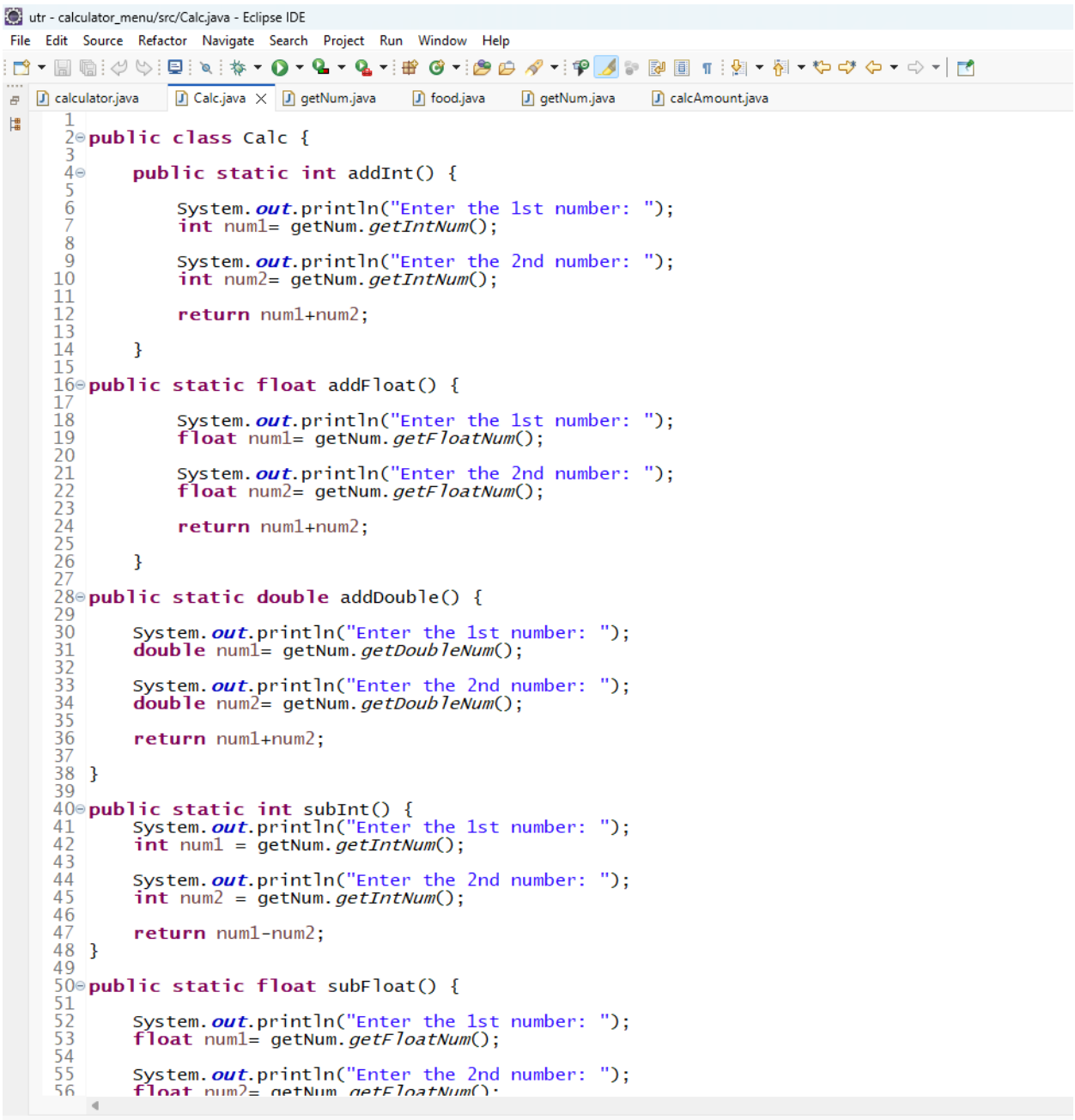
            //substraction of integers
            case 4:

                System.out.println("Substraction of two integer numbers are: "+ Calc.subInt());
                break;

            //substraction of float
```



File:- Calc.java




```java
public class Calc {

    public static int addInt() {

        System.out.println("Enter the 1st number: ");
        int num1= getNum.getIntNum();

        System.out.println("Enter the 2nd number: ");
        int num2= getNum.getIntNum();

        return num1+num2;

    }

public static float addFloat() {

        System.out.println("Enter the 1st number: ");
        float num1= getNum.getFloatNum();

        System.out.println("Enter the 2nd number: ");
        float num2= getNum.getFloatNum();

        return num1+num2;

    }

public static double addDouble() {

    System.out.println("Enter the 1st number: ");
    double num1= getNum.getDoubleNum();

    System.out.println("Enter the 2nd number: ");
    double num2= getNum.getDoubleNum();

    return num1+num2;

}

public static int subInt() {
    System.out.println("Enter the 1st number: ");
    int num1 = getNum.getIntNum();

    System.out.println("Enter the 2nd number: ");
    int num2 = getNum.getIntNum();

    return num1-num2;
}

public static float subFloat() {

    System.out.println("Enter the 1st number: ");
    float num1= getNum.getFloatNum();

    System.out.println("Enter the 2nd number: ");
    float num2= getNum.getFloatNum();
```

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```
55      System.out.println("Enter the 2nd number: ");
56      float num2= getNum.getFloatNum();
57
58      return num1-num2;
59
60  }
61
62  public static double subDouble() {
63
64      System.out.println("Enter the 1st number: ");
65      double num1= getNum.getDoubleNum();
66
67      System.out.println("Enter the 2nd number: ");
68      double num2= getNum.getDoubleNum();
69
70      return num1-num2;
71
72  }
73
74  public static int mulInt() {
75      System.out.println("Enter the 1st number: ");
76      int num1 = getNum.getIntNum();
77
78      System.out.println("Enter the 2nd number: ");
79      int num2 = getNum.getIntNum();
80
81      return num1*num2;
82  }
83
84  public static float mulFloat() {
85
86      System.out.println("Enter the 1st number: ");
87      float num1= getNum.getFloatNum();
88
89      System.out.println("Enter the 2nd number: ");
90      float num2= getNum.getFloatNum();
91
92      return num1*num2;
93
94  }
95
96  public static double mulDouble() {
97
98      System.out.println("Enter the 1st number: ");
99      double num1= getNum.getDoubleNum();
100
101     System.out.println("Enter the 2nd number: ");
102     double num2= getNum.getDoubleNum();
103
104     return num1*num2;
105
106 }
107
108 public static int DivInt() {
109     System.out.println("Enter the 1st number: ");
110     int num1 = getNum.getIntNum();
```

```
108 public static int DivInt() {
109     System.out.println("Enter the 1st number: ");
110     int num1 = getNum.getIntNum();
111
112     System.out.println("Enter the 2nd number: ");
113     int num2 = getNum.getIntNum();
114
115     return num1/num2;
116 }
117
118 public static float DivFloat() {
119
120     System.out.println("Enter the 1st number: ");
121     float num1= getNum.getFloatNum();
122
123     System.out.println("Enter the 2nd number: ");
124     float num2= getNum.getFloatNum();
125
126     return num1/num2;
127
128 }
129
130 public static double DivDouble() {
131
132     System.out.println("Enter the 1st number: ");
133     double num1= getNum.getDoubleNum();
134
135     System.out.println("Enter the 2nd number: ");
136     double num2= getNum.getDoubleNum();
137
138     return num1/num2;
139
140 }
141
142 }
143
```

File:- getNum.java

calculator.java | Calc.java | getNum.java | food.java | getNum.java | calcAmount.java

```java
public class getNum {

    public static int getchoice() {
        try {

        byte [] choiceInput=new byte[100];

        int length= System.in.read(choiceInput);

        byte [] newChoice_input=new byte[length-2];

        System.arraycopy(choiceInput, 0, newChoice_input, 0, length-2);

        String s=new String(newChoice_input);

        return  Integer.parseInt(s);
        } catch(Exception e) {
            System.out.println(e);
        }
        return -1;
    }

    public static int getIntNum() {
        try {

        byte [] choiceInput=new byte[100];

        int length= System.in.read(choiceInput);

        byte [] newChoice_input=new byte[length-2];

        System.arraycopy(choiceInput, 0, newChoice_input, 0, length-2);

        String s=new String(newChoice_input);

        return  Integer.parseInt(s);
        } catch(Exception e) {
            System.out.println(e);
        }
        return -1;
    }

    public static float getFloatNum() {
        try {

        byte [] choiceInput=new byte[100];

        int length= System.in.read(choiceInput);

        byte [] newChoice_input=new byte[length-2];

        System.arraycopy(choiceInput, 0, newChoice_input, 0, length-2);

        String s=new String(newChoice_input);
```

```
        System.arraycopy(choiceInput, 0, newChoice_input, 0, length-2);

        String s=new String(newChoice_input);

        return  Float.parseFloat(s);
        } catch(Exception e) {
            System.out.println(e);
        }
        return -1;
    }

    public static double getDoubleNum() {
        try {

        byte [] choiceInput=new byte[100];

        int length= System.in.read(choiceInput);

        byte [] newChoice_input=new byte[length-2];

        System.arraycopy(choiceInput, 0, newChoice_input, 0, length-2);

        String s=new String(newChoice_input);

        return  Double.parseDouble(s);
        } catch(Exception e) {
            System.out.println(e);
        }
        return -1;
    }

}
```

Que2:-

Display food menu to user. User will select items from menu along with the quantity. (eg 1. Dosa 2. Samosa 3. Idli ... 10 . Generate Bill) Assign fixed prices to food items(hard code the prices)

When user enters 'Generate Bill' option(10) , display total bill & exit.

File:- food.java

calculator.java | Calc.java | getNum.java | food.java | getNum.java | calcAmount.java

```java
public class food {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        int i =0;
        int Total_Amount=0,qty,price;
        int choice;
        String items[]=new String[100];
        int items_price[]=new int[100];
        int quantity[]=new int[100];
        do {
            System.out.println("====================MENU================");
            System.out.println("1. Samosa:");
            System.out.println("2. Poha (per plate):");
            System.out.println("3. Jalebi (250 gm):");
            System.out.println("4. Idli:");
            System.out.println("5. Dosa:");
            System.out.println("6. Butter Chicken : ");
            System.out.println("7. Aalo Paratha :");
            System.out.println("8. Mutton Korma :");
            System.out.println("9. Butter Paneer Masala :");
            System.out.println("10.Total Bill & Exit :");
            System.out.println("=======================================");



            System.out.println("Enter your choice");
             choice=getNum.getchoice();

            switch(choice) {

            //addition of integers
            case 1:

                System.out.println("Enter quantity: ");
                 qty=getNum.getchoice();
                 price=20;
                 Total_Amount+=calcAmount.Amount(qty, price);
                items[i]="Samosa";
                items_price[i]=price;
                quantity[i]=qty;
                i++;
                break;

            //additon of two float

            case 2:

                System.out.println("Enter quantity: ");
                 qty=getNum.getchoice();
                 price=35;
                Total_Amount+=calcAmount.Amount(qty, price);
                items[i]="Poha (per plate)";
                items_price[i]=price;
```

```
56                  quantity[i]=qty;
57                  i++;
58
59                  break;
60
61              //addition of two double
62              case 3:
63                  System.out.println("Enter quantity: ");
64                   qty=getNum.getchoice();
65                   price=80;
66                  Total_Amount+=calcAmount.Amount(qty, price);
67                  items[i]="Jalebi (250 gm)";
68                  items_price[i]=price;
69                  quantity[i]=qty;
70                  i++;
71
72                  break;
73
74              //substraction of integers
75              case 4:
76
77                  System.out.println("Enter quantity: ");
78                   qty=getNum.getchoice();
79                   price=50;
80                  Total_Amount+=calcAmount.Amount(qty, price);
81                  items[i]="Idli";
82                  items_price[i]=price;
83                  quantity[i]=qty;
84                  i++;
85                  break;
86
87              //substraction of float
88              case 5:
89
90                  System.out.println("Enter quantity: ");
91                   qty=getNum.getchoice();
92                   price=40;
93                  Total_Amount+=calcAmount.Amount(qty, price);
94                  items[i]="Dosa";
95                  items_price[i]=price;
96                  quantity[i]=qty;
97                  i++;
98                      break;
99
100             //substraction of Double
101             case 6:
102
103                 System.out.println("Enter quantity 1. Full 2. Half : ");
104                  qty=getNum.getchoice();
105                  if(qty==1) {
106                      price=550;
107                         Total_Amount+=calcAmount.Amount(qty, price);
108                         items[i]="Butter Chicken";
109                         items_price[i]=price;
110                         quantity[i]=qty;
```

calculator.java | Calc.java | getNum.java | food.java | getNum.java | calcAmount.java

```
                            items_price[i]=price;
                            quantity[i]=qty;
                            i++;
                     }
                     else {
                     price=300;
                    Total_Amount+=calcAmount.Amount(qty, price);
                    items[i]="Butter Chicken";
                    items_price[i]=price;
                    quantity[i]=qty;
                    i++;
                    }

                        break;

                //Multiplication of integers
                case 7:

                    System.out.println("Enter quantity: ");
                     qty=getNum.getchoice();
                     price=60;
                    Total_Amount+=calcAmount.Amount(qty, price);
                    items[i]="Aalo Paratha";
                    items_price[i]=price;
                    quantity[i]=qty;
                    i++;
                        break;

                //Multiplication of float
                case 8:

                    System.out.println("Enter quantity 1. Full 2. Half : ");
                     qty=getNum.getchoice();
                     if(qty==1) {
                         price=850;
                            Total_Amount+=calcAmount.Amount(qty, price);
                            items[i]="Mutton Korma";
                            items_price[i]=price;
                            quantity[i]=qty;
                            i++;

                     }
                     else
                     {
                     price=450;
                    Total_Amount+=calcAmount.Amount(qty, price);
                    items[i]="Mutton Korma";
                    items_price[i]=price;
                    quantity[i]=qty;
                    i++;
                     }
                        break;

               //Multiplication of doubles
                case 9:
```

```
                }
                    break;

            //Multiplication of doubles
             case 9:

                 System.out.println("Enter quantity: ");
                  qty=getNum.getchoice();
                  price=180;
                 Total_Amount+=calcAmount.Amount(qty, price);
                 items[i]="Butter Paneer Masala";
                 items_price[i]=price;
                 quantity[i]=qty;
                 i++;
                    break;

//
             case 10:

                 System.out.println("<==================Generating the bills===============>");
                 System.out.println("Items Name "+"              Price"+"              Quantity");
                 for(int j=0;j<i;j++) {
                     System.out.println(items[j]+"                  "+ items_price[j]+"                  "+ quantity[j]);
                 }

                 System.out.println("Total amount to be paid: "+ Total_Amount);

                      return;

             default:

                     System.out.println("Enter the valid choice");
                 }

            }while(choice!=10);
    }



    }
```

File:- calcAmount.java

calculator.java | Calc.java | getNum.java | food.java | getNum.java | calcAmount.java

```
1
2  public class calcAmount {
3
4      public static int Amount(int qty, int p) {
5
6          return qty*p;
7      }
8
9  }
10
```

File:-getNum.java

calculator.java | Calc.java | getNum.java | food.java | getNum.java | calcAmount.java

```java
public class getNum {

    public static int getchoice() {

    try {

        byte [] choiceInput=new byte[100];

        int length= System.in.read(choiceInput);

        byte [] newChoice_input=new byte[length-2];

        System.arraycopy(choiceInput, 0, newChoice_input, 0, length-2);

        String s=new String(newChoice_input);

        return  Integer.parseInt(s);
        } catch(Exception e) {
            System.out.println(e);
        }
        return -1;
    }



}
```

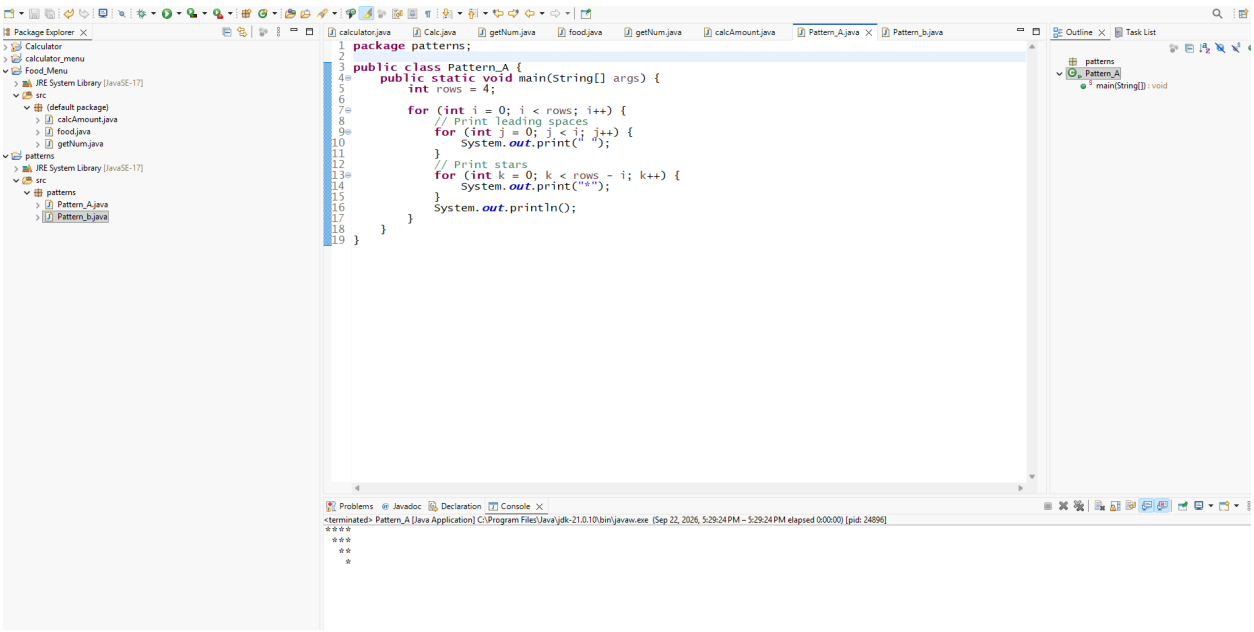
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Package Explorer
Calculator
calculator_menu
Food_Menu
JRE System Library [JavaSE-17]
src
(default package)
calcAmount.java
food.java
getNum.java
patterns
JRE System Library [JavaSE-17]
src
patterns
Pattern_A.java
Pattern_b.java
calculator.java
Calc.java
getNum.java
food.java
getNum.java
calcAmount.java
Pattern_A.java
Pattern_b.java
1 package patterns;
2
3 public class Pattern_A {
4 public static void main(String[] args) {
5 int rows = 4;
6
7 for (int i = 0; i < rows; i++) {
8 // Print leading spaces
9 for (int j = 0; j < i; j++) {
10 System.out.print(" ");
11 }
12 // Print stars
13 for (int k = 0; k < rows - i; k++) {
14 System.out.print("*");
15 }
16 System.out.println();
17 }
18 }
19 }
Outline
Task List
patterns
Pattern_A
main(String[]) : void
Problems
Javadoc
Declaration
Console
<terminated> Pattern_A [Java Application] C:\Program Files\Java\jdk-21.0.10\bin\javaw.exe (Sep 22, 2026, 5:29:24 PM – 5:29:24 PM elapsed 0:00:00) [pid: 24896]
****
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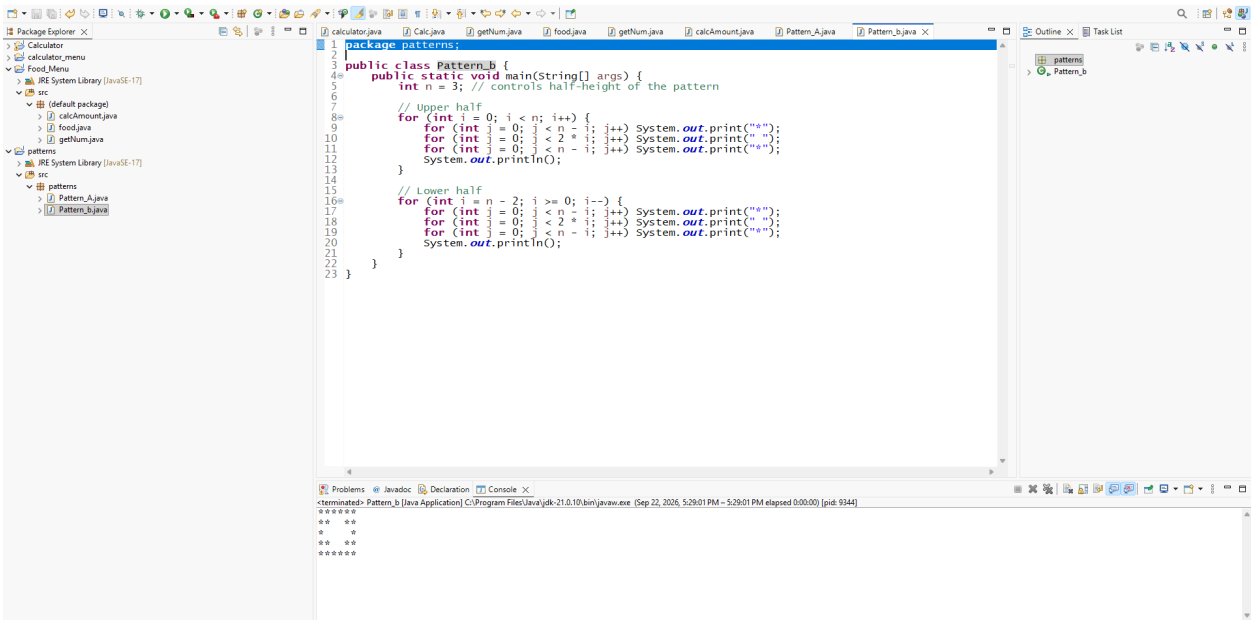
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Package Explorer
Calculator
calculator_menu
Food_Menu
JRE System Library [JavaSE-17]
src
(default package)
calcAmount.java
food.java
getNum.java
patterns
JRE System Library [JavaSE-17]
src
patterns
Pattern_A.java
Pattern_b.java
calculator.java
Calc.java
getNum.java
food.java
getNum.java
calcAmount.java
Pattern_A.java
Pattern_b.java
1 package patterns;
2
3 public class Pattern_b {
4 public static void main(String[] args) {
5 int n = 3; // controls half-height of the pattern
6
7 // Upper half
8 for (int i = 0; i < n; i++) {
9 for (int j = 0; j < n - i; j++) System.out.print("*");
10 for (int j = 0; j < 2 * i; j++) System.out.print(" ");
11 for (int j = 0; j < n - i; j++) System.out.print("*");
12 System.out.println();
13 }
14
15 // Lower half
16 for (int i = n - 2; i >= 0; i--) {
17 for (int j = 0; j < n - i; j++) System.out.print("*");
18 for (int j = 0; j < 2 * i; j++) System.out.print(" ");
19 for (int j = 0; j < n - i; j++) System.out.print("*");
20 System.out.println();
21 }
22 }
23 }
Outline
Task List
patterns
Pattern_b
Problems
Javadoc
Declaration
Console
<terminated> Pattern_b [Java Application] C:\Program Files\Java\jdk-21.0.10\bin\javaw.exe (Sep 22, 2026, 5:29:01 PM – 5:29:01 PM elapsed 0:00:00) [pid: 9344]
******
** **
* *
** **
******